Instability as a Data Quality Issue in Wikipedia
Example of women-related biography pages

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KODAQS Cohort 2; Assignment

Wikipedia as a Dynamic Data System

- Wikipedia articles are continuously revised over time
- Revision history reveals temporal variation in data quality

Research Gap

What prior research focuses on

- Content quality (accuracy, bias)
- Revision metrics (edits, reverts)

What is missing

- Topic-specific instability

  e.g., women-related pages

Research Questions and Hypothesis

RQ1 Are gender-related pages more unstable?

RQ2 What type of instability ?

1) **(structural instability)**: higher edit concentration
2) **(temporal instability)**: higher burst-day rates
3) **(revert-related instability)**: higher reverted edit share

RQ3 How can instability be mitigated?

Data : English Wikipedia

Key Definition

gender-related pages: Label: *Culture.Biography.Women*

Types of Instability

1) Structural instability

→ concentration of a page's edits on its single most active day

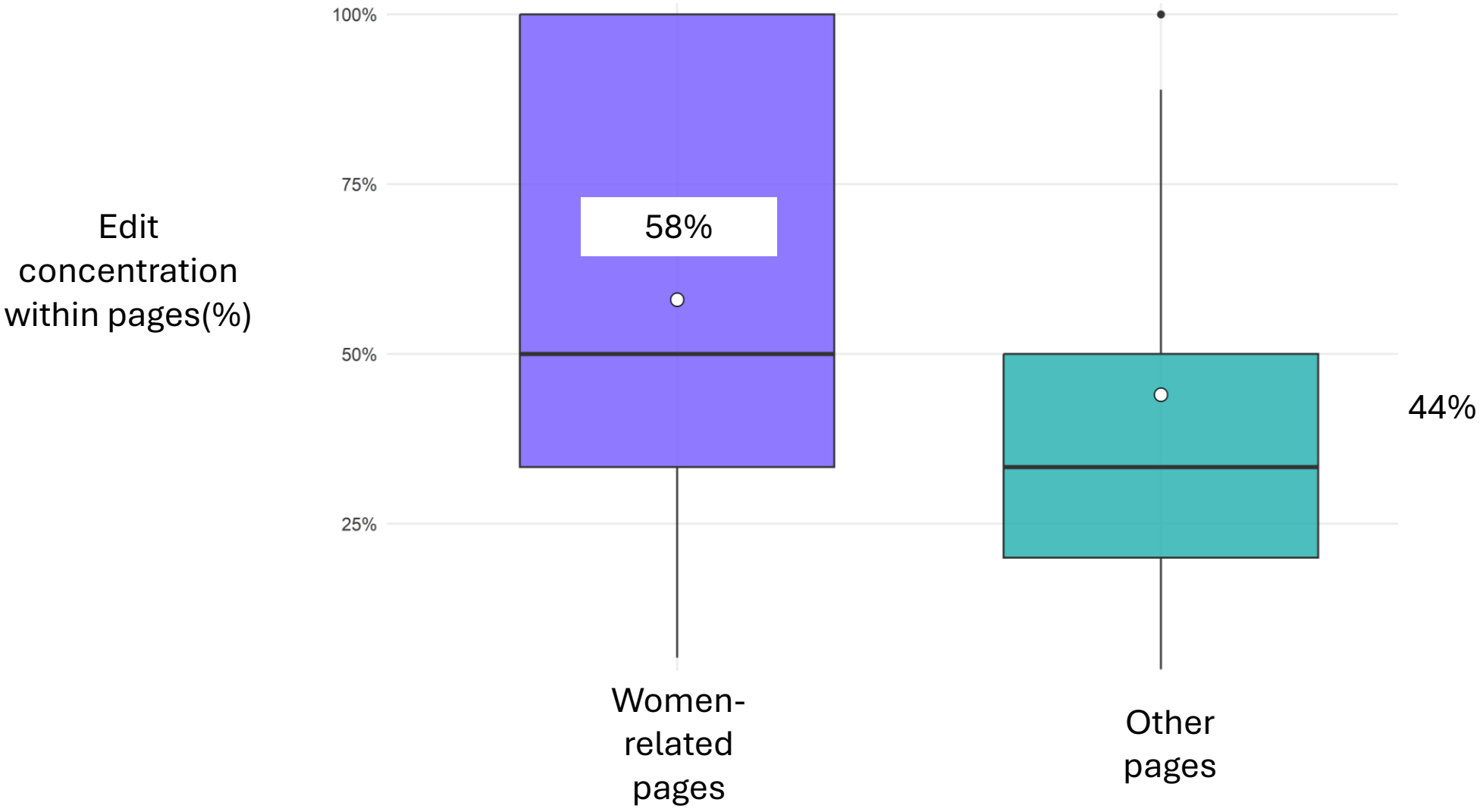
2) Temporal instability

→ share of active days on a page that are classified as burst days relative to that page's own edit distribution

3) Revert-related instability

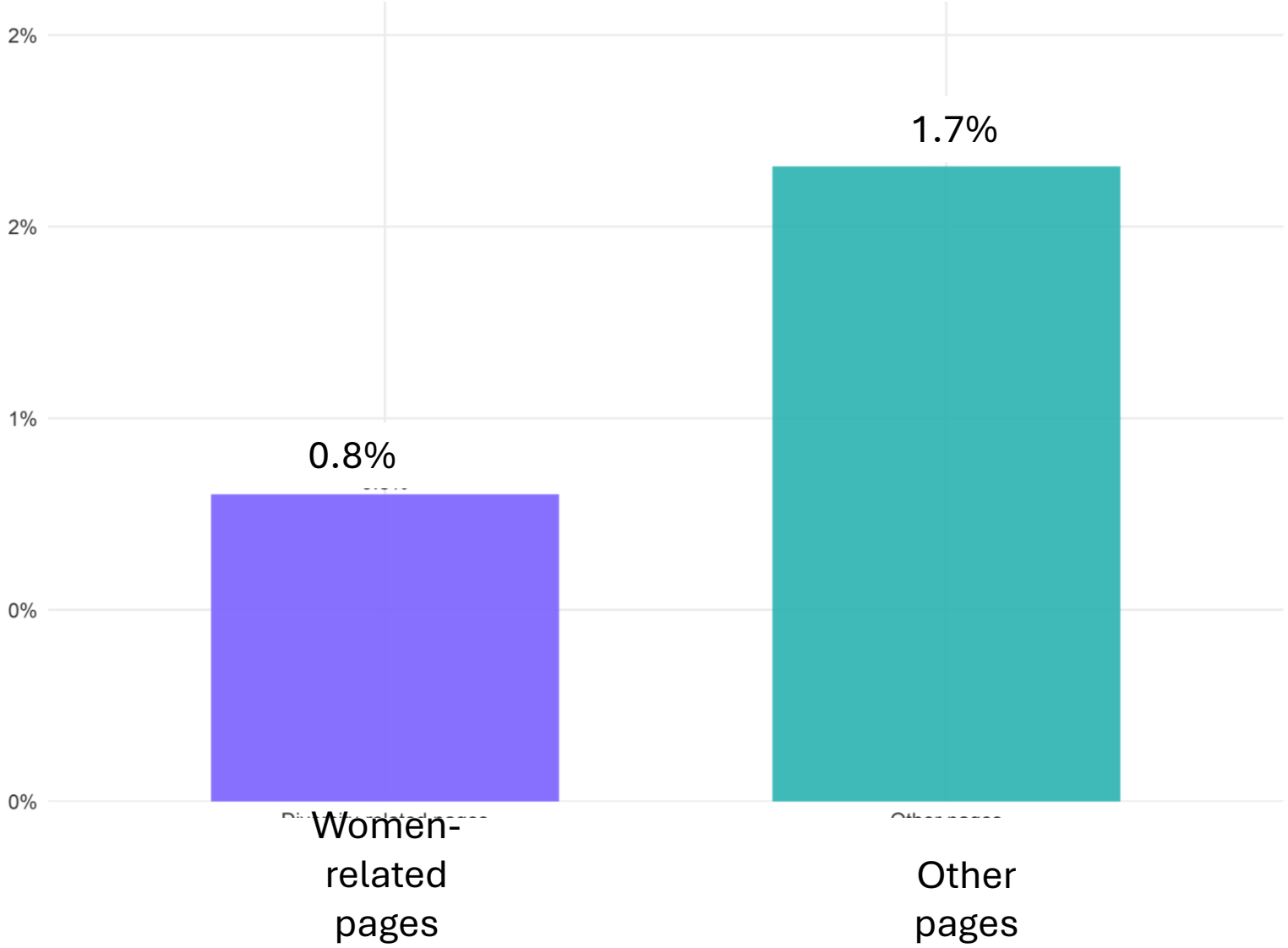
→ share of edits on a page that are reverted

1. Structural Instability is higher for diversity-related pages

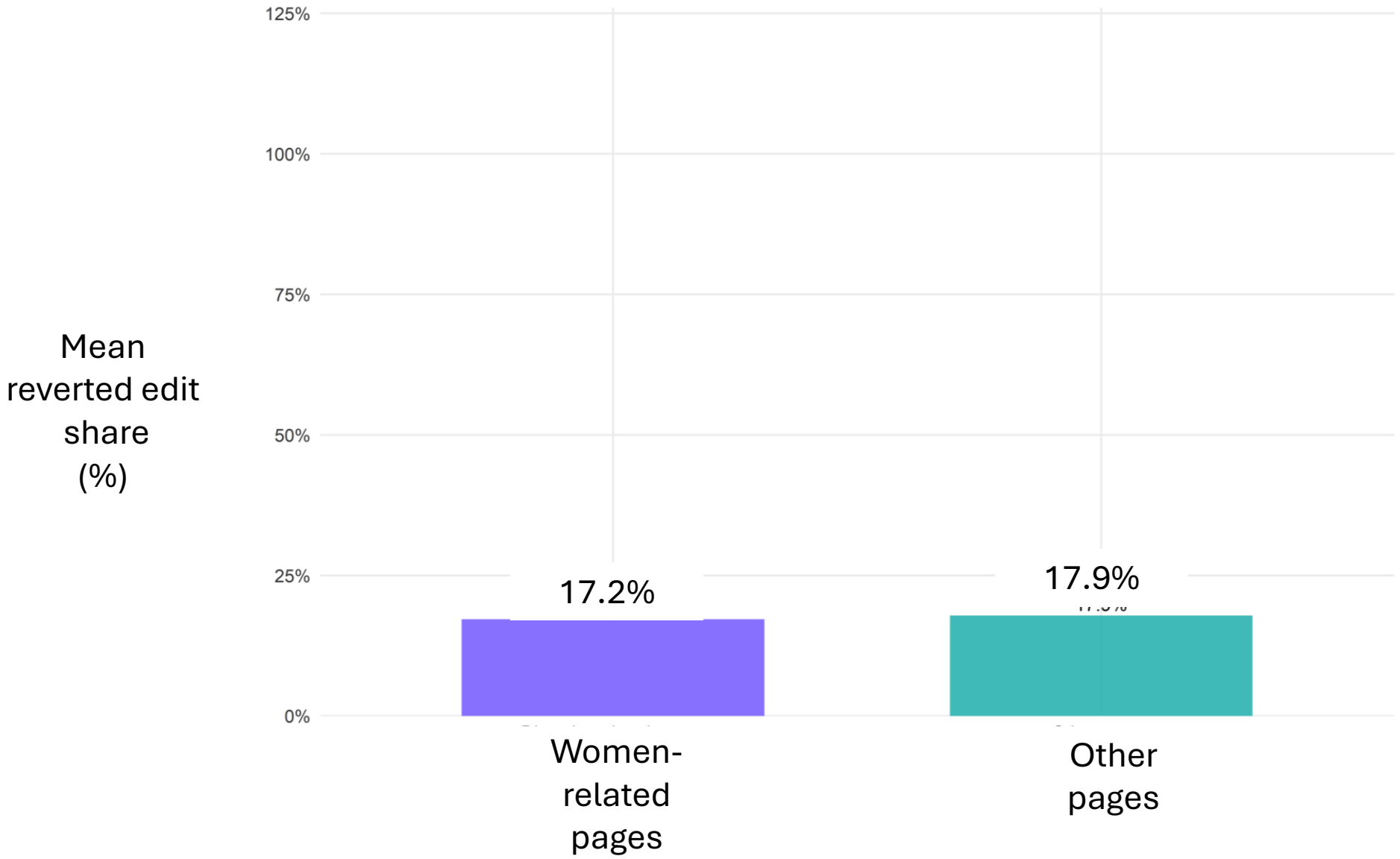


2. Temporal stability is higher for women-related pages

Mean burst-day Rate within page(%)



3. Revert-related instability does not differ by topic



Mitigation Strategies for Structural Instability

Compare 3 strategies

1. Burst weighting

→ Downweight pages with highly concentrated edits (high-burst pages have less influence)

2. Event-day trimming

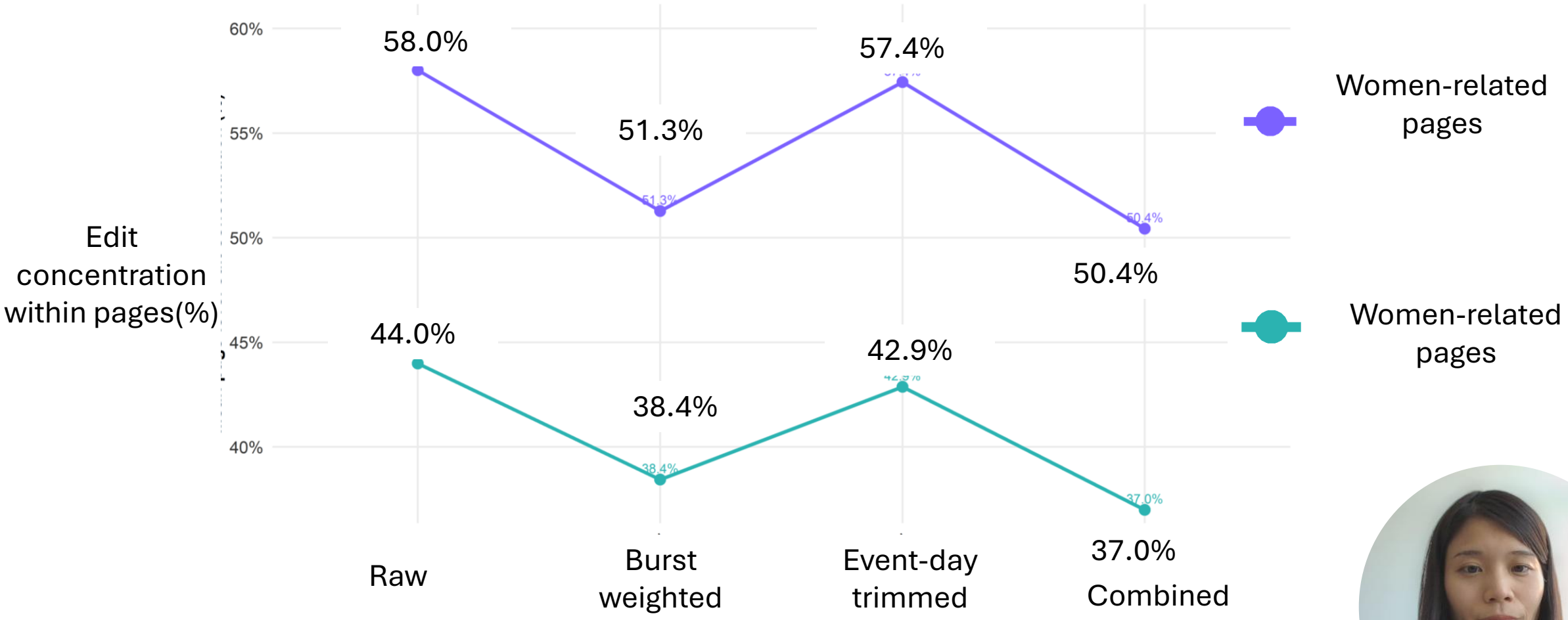
→ Remove unusually intense editing days (recompute metrics without burst days)

3. Combined approach

→ Remove burst days and downweight bursty pages

Then, evaluate their effects on **structural instability**

Mitigation Strategy on Structural Instability



Summary

RQ1: Are diversity-related pages more unstable?
→ Yes, but only in terms of structural instability

RQ2: What drives instability in this case(diversity VS others)?
→ Temporal concentration, not editorial conflict
→ No difference in revert-related instability

RQ3: Can instability be mitigated?
→ Yes, through burst-focused strategies and combined design.
→ But the gap between topic groups persists.

Key implication
Rather than focusing on a single metric,

we should evaluate a variety of metrics.